

CHY1008

Chemistry and Environmental Studies

Module 2

Part 2

States of Matter, Thermodynamics, Kinetics and Catalysis

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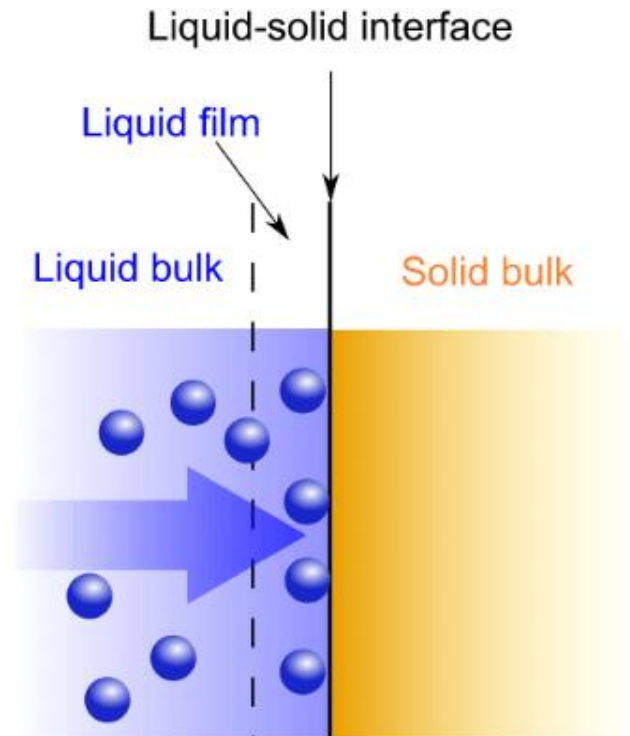
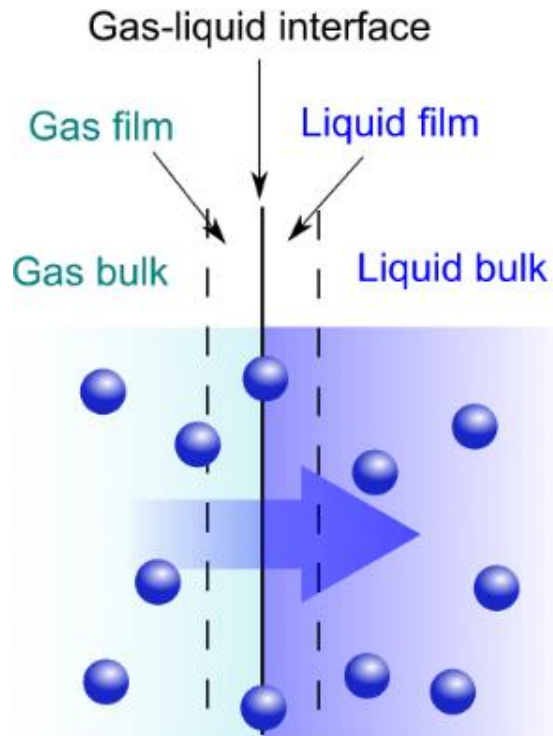
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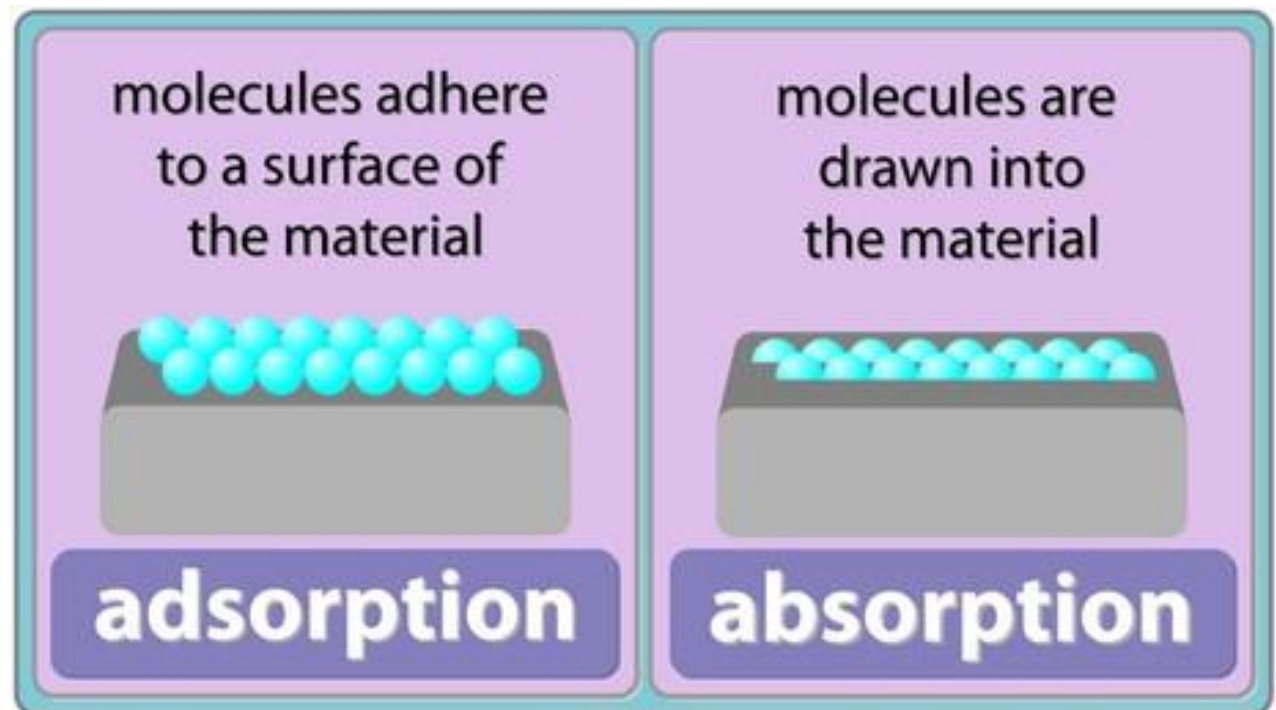
Solid-Gas Interface

Surface Chemistry



Solid Gas Interface

Surface Chemistry



Solid Gas Interface

Surface Chemistry

- **Adsorption:** is the adhesion of atoms, ions, biomolecules or molecules of gas, liquid, or dissolved solids to a surface.
- **Absorption:** is a physical or chemical phenomenon or a process in which atoms, molecules, or ions enter some bulk phase - gas, liquid, or solid material.

Adsorption

Accumulation of the molecular species at the surface rather than in the bulk of the solid or liquid is termed as adsorption.

It is a surface phenomenon.

Exothermic process

It is favored by low temperature

It steadily increases and reaches equilibrium.

Concentration on the surface of adsorbent is different from that in the bulk.

Absorption

Assimilation of molecular species throughout the bulk of the solid or liquid is termed as absorption.

It is a bulk phenomenon

Endothermic process

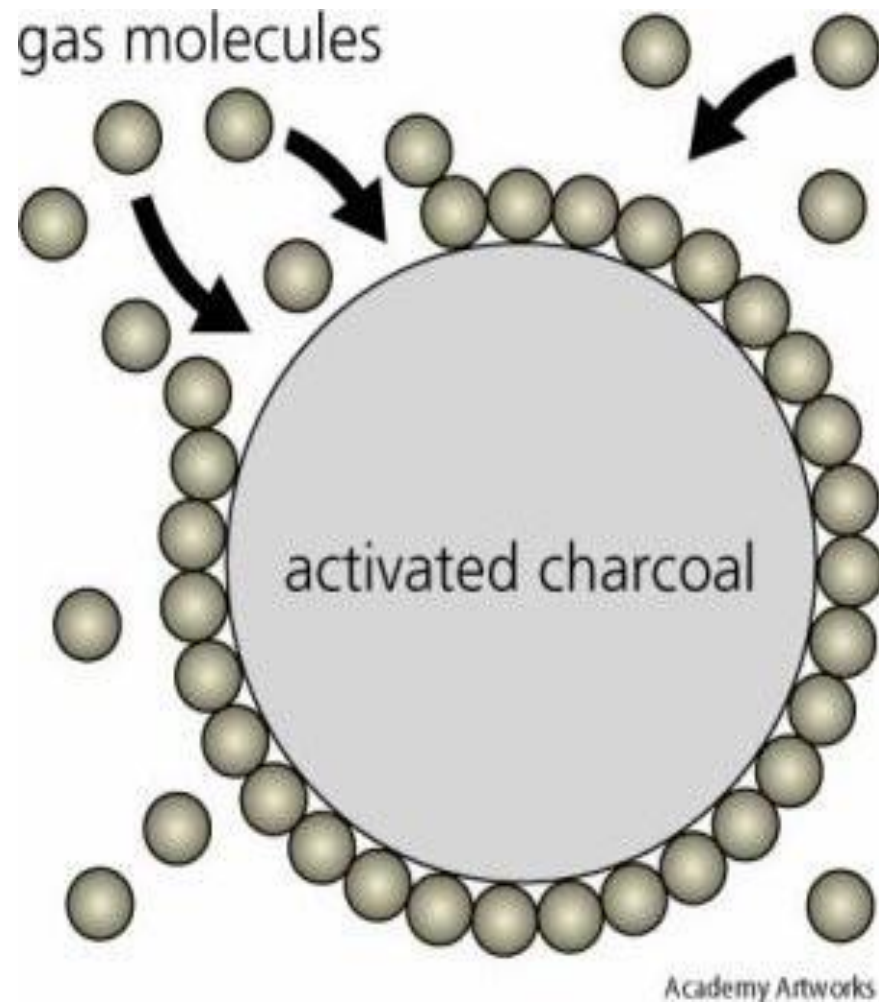
It is not affected by temperature

It occurs at a uniform rate.

It is same throughout the material.

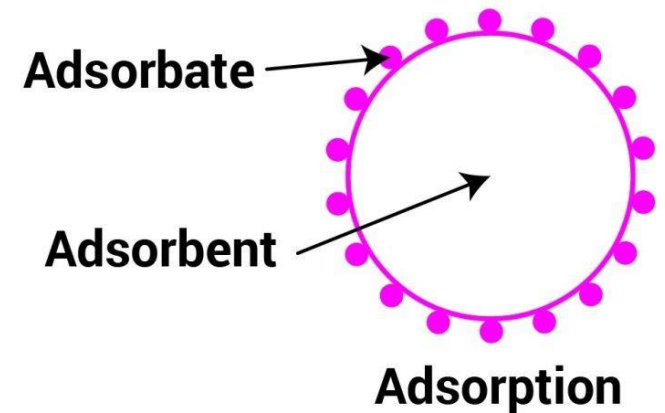
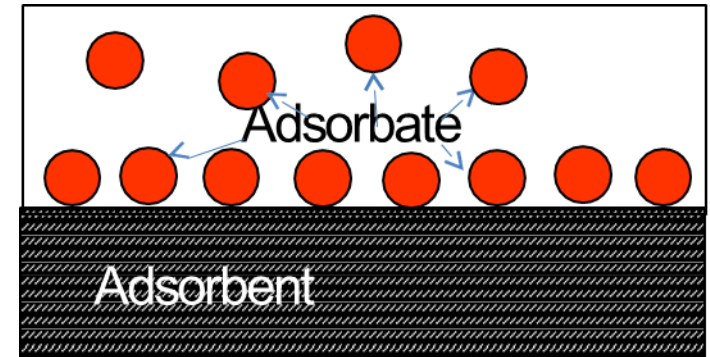
Adsorption

- Adsorption is the adhesion of atoms, ions or molecules from a gas, liquid or dissolved solid to a surface.
- It is a surface phenomenon.



Adsorption

- **Adsorbate:** material being adsorbed.
- **Adsorbent:** material doing the adsorbing. (examples are activated carbon or ion exchange resin)

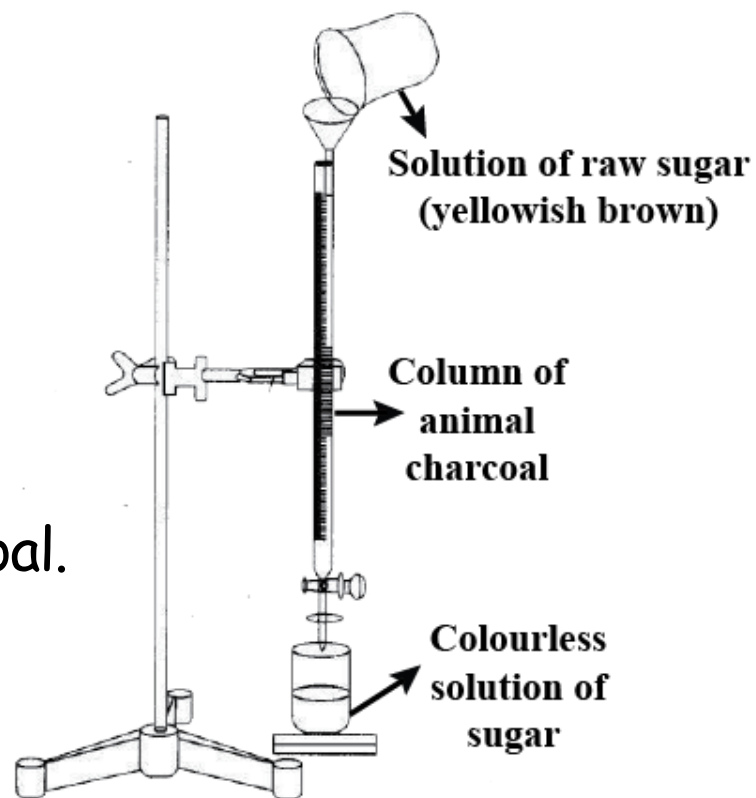


Adsorption

- ♦ The accumulation of molecular species at the surface rather than in the bulk of a solid or liquid is termed adsorption.
- ♦ The molecular species or substance, which concentrates or accumulates at the surface is termed adsorbate and the material on the surface of which the adsorption takes place is called adsorbent

Examples of Adsorption

- ♦ Adsorption of O_2 , H_2 , CO , Cl_2 , NH_3 or SO_2 on surface of charcoal,
- ♦ In a solution of an organic dye, say **methylene blue** on the surface of **charcoal**, i.e., are adsorbed.
- ♦ Aqueous solution of **raw sugar**, when passed over beds of animal charcoal, becomes colourless as the colouring substances are adsorbed by the charcoal.

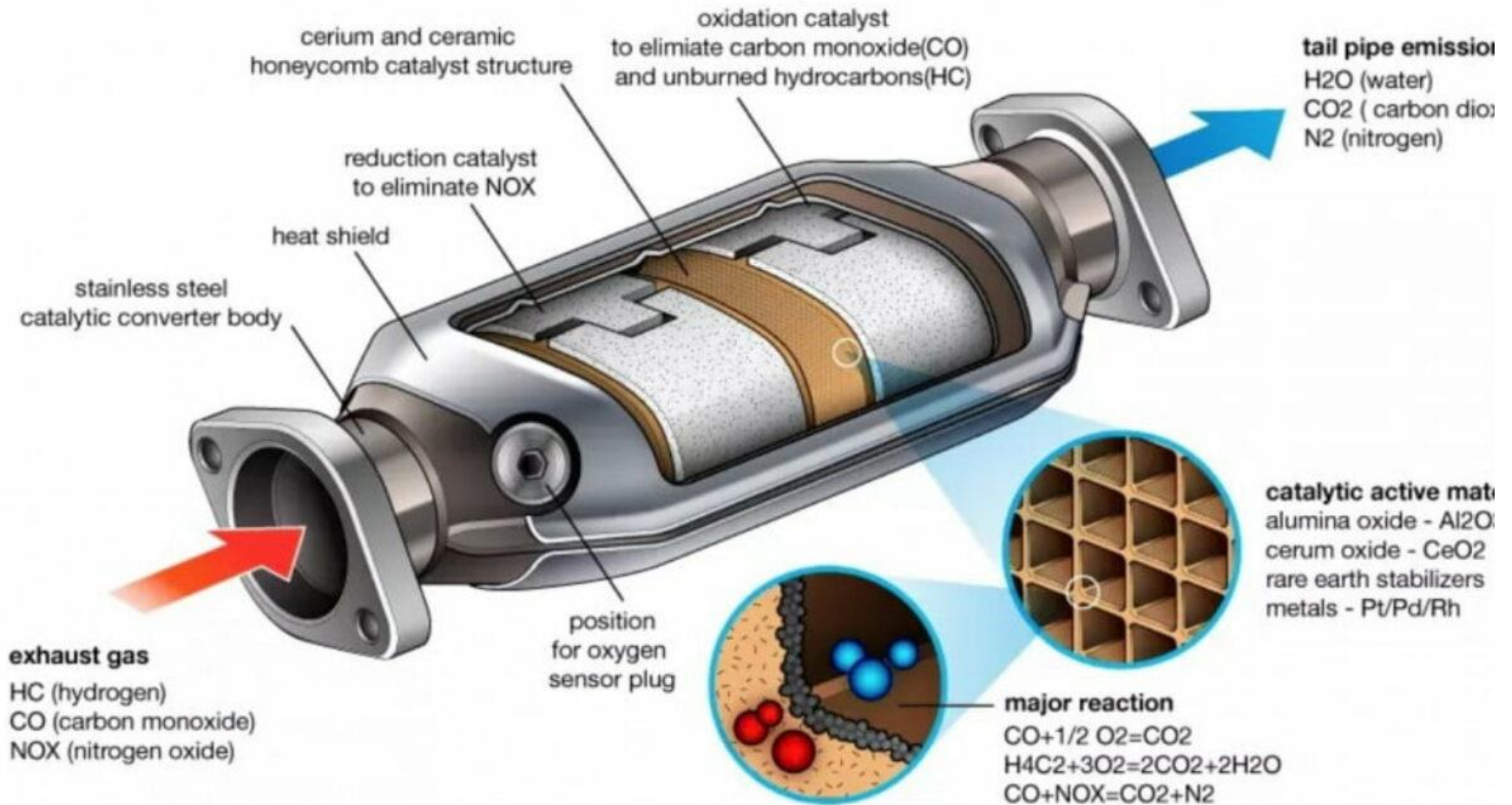


Examples of Adsorption

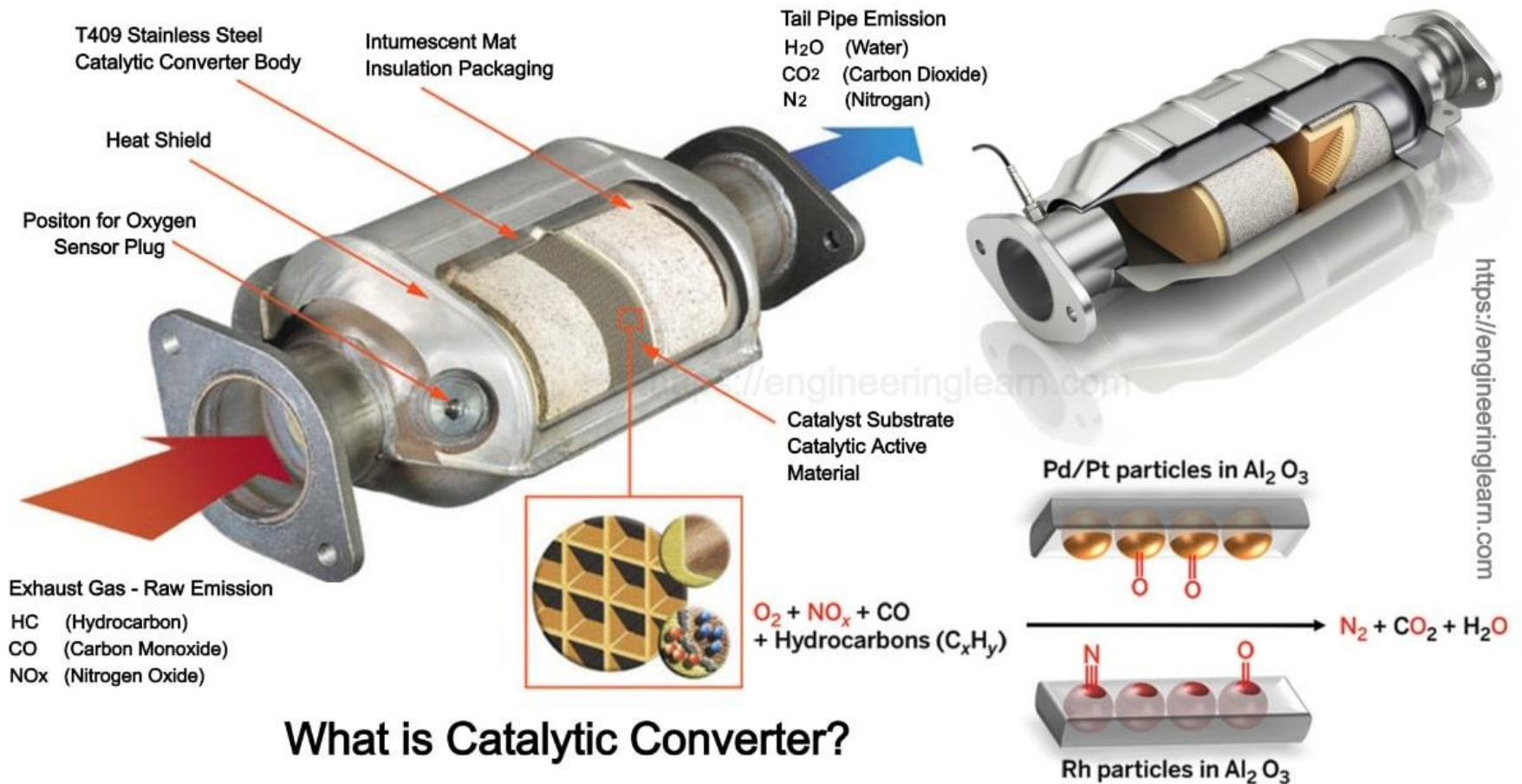
- ♦ The air becomes dry in the presence of **silica gel** because the water molecules get adsorbed on the surface of the gel.



Catalytic Converters



Platinum and Rhodium

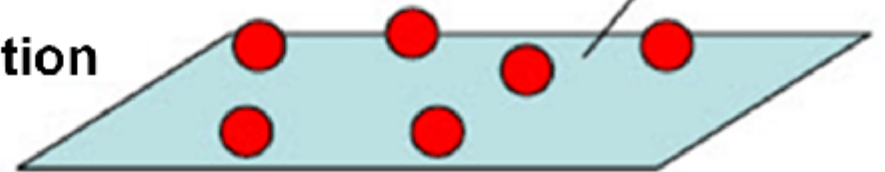


Type of Adsorption

(a) Adsorption

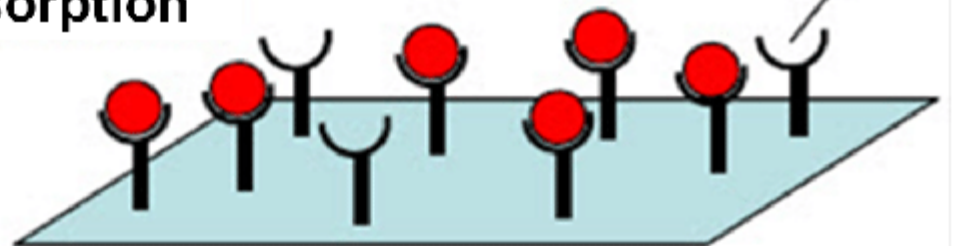
Physisorption

Substrate Surface



Chemisorption

Binding site



Type of Adsorption

Physisorption

- physical adsorption is a type of adsorption in which the adsorbate adheres to the surface only through Van der Waals (weak intermolecular) interactions, which are also responsible for the non-ideal behavior of real gases.

Chemisorption or chemical adsorption

- The gas molecules or atoms are held to the solid by chemical bonds. For example, hydrogen is chemisorbed on nickel

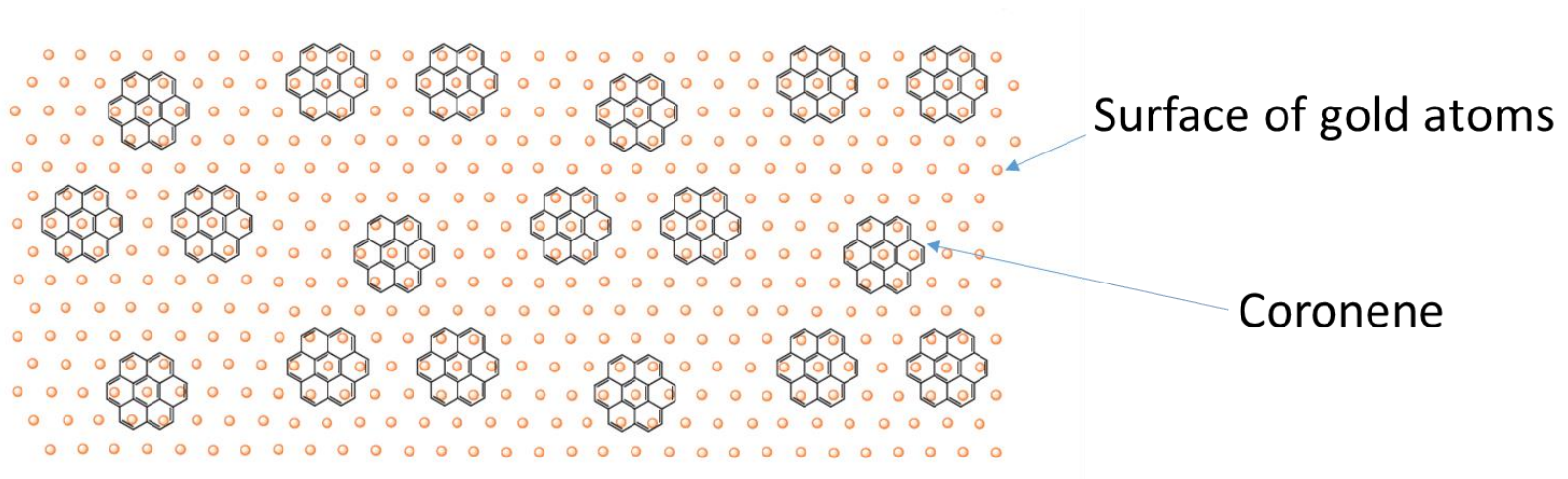
- **Long range interaction:** Physical contact, due to Van der Waals force.

Physisorption

- **Short range interaction:** Chemical bonding, such as covalent, ionic, metallic bonding

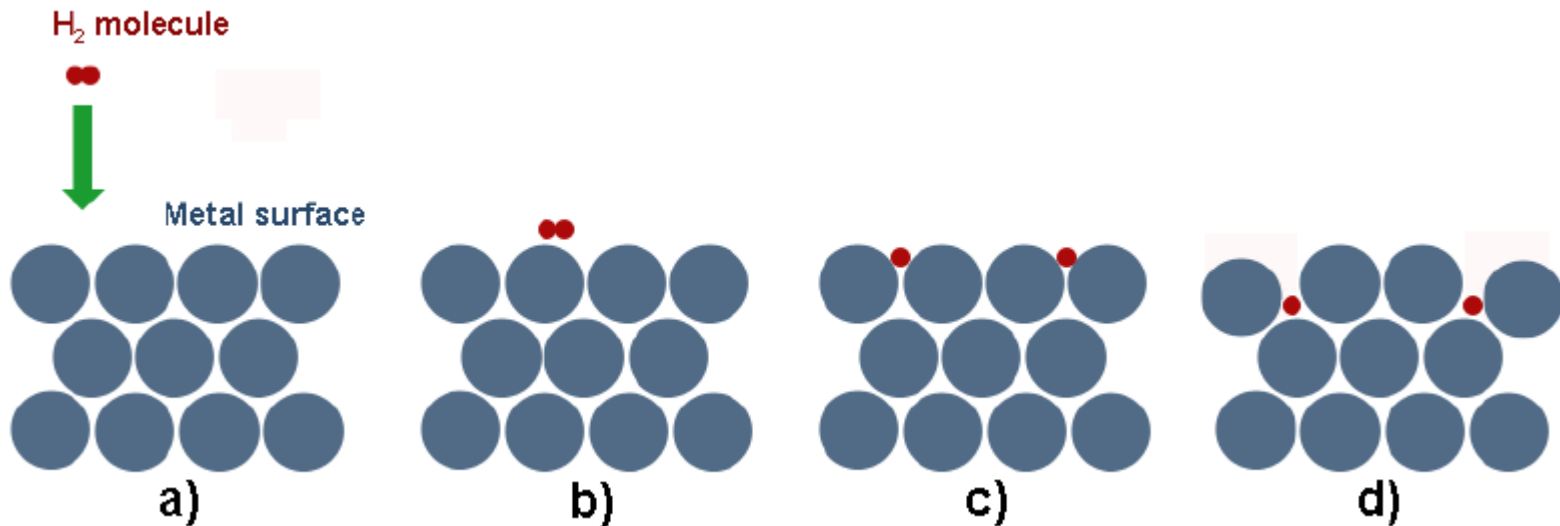
Chemisorption

Example of Physisorption



Sticking of benzene-like molecules, for example coronene, on the surface of gold (111) is an example of physisorption.

Example of Chemisorption



- a) Metal surface (adsorbent)
- b) Physisorption of H₂ through van der Waals interaction
- c) Chemisorptions of H₂ through M-H bond (metal hydrides)
- d) Chemisorbed H atom may slid into the bulk metal through diffusion

“Chemisorption always accompanied by physisorption”

Thermodynamic Considerations

- Adsorption (or any) spontaneous process requires $\Delta G < 0$. Because the translational freedom of the adsorbate is reduced when it is adsorbed, **ΔS is negative**.
- Therefore, in order for **$\Delta G = \Delta H - T\Delta S$** to be negative, ΔH must be negative (that is, the process is exothermic).
- Exceptions may occur if the adsorbate dissociates and has high translational mobility on the surface.

Thermodynamic consideration

A spontaneous process requires $\Delta G < 0$.

ΔS is negative for the adsorbate

Therefore, from

$$\Delta G = \Delta H - T\Delta S$$

In order for ΔG be negative

ΔH must be negative

i.e. the process should be **exothermic** and
**the degree of adsorption increases with
decreasing temperature**

Enthalpies of Physi- and Chemisorption

Synoptic table 25.1* Maximum observed enthalpies of physisorption

Adsorbate	$\Delta_{\text{ad}}H^\circ/(\text{kJ mol}^{-1})$
CH ₄	-21
H ₂	-84
H ₂ O	-59
N ₂	-21

* More values are given in the *Data section*.

Synoptic table 25.2* Enthalpies of chemisorption, $\Delta_{\text{ad}}H^\circ/(\text{kJ mol}^{-1})$

Adsorbate	Adsorbent (substrate)		
	Cr	Fe	Ni
CH ₄	-427	-285	-243
CO		-192	
H ₂	-188	-134	
NH ₃		-188	-155

* More values are given in the *Data section*.

Exothermic reactions, enthalpy change is negative

Enthalpies of *chemisorption are greater than enthalpies of physisorption*. These enthalpies are close to covalent bond enthalpies.

Physisorption

WEAK, LONG RANGE BONDING

Van der Waals interactions (e.g. London dispersion, dipole-dipole), hydrogen bonding.

NOT SURFACE SPECIFIC

Physisorption takes place between all molecules on any surface providing the temperature is low enough.

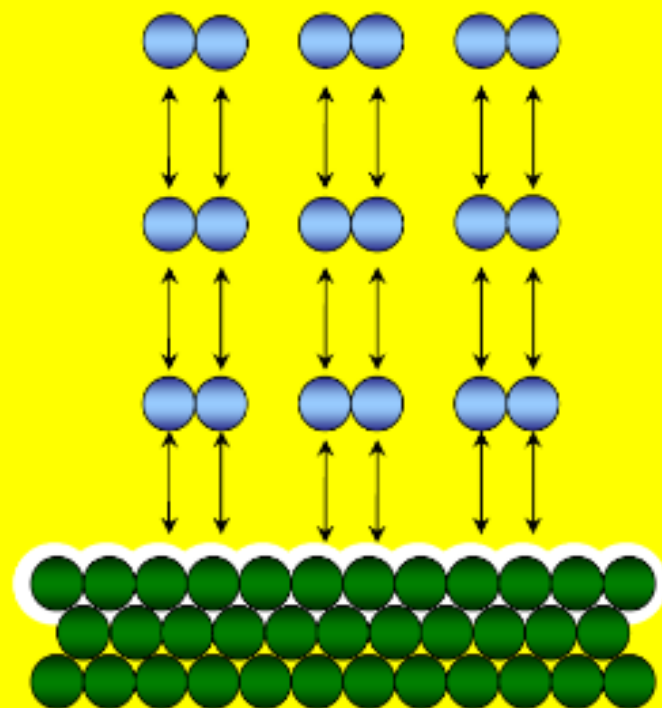
$$\Delta H_{\text{ads}} = 5\text{--}50 \text{ kJ/mol}$$

Non activated with equilibrium achieved relatively quickly.
Increasing temperature always reduces surface coverage.

No surface reactions.

MULTILAYER ADSORPTION

BET isotherm used to model adsorption equilibrium



Chemisorption

STRONG, SHORT RANGE BONDING

Chemical bonding involving orbital overlap and charge transfer

SURFACE SPECIFIC

e.g. chemisorption of hydrogen takes place on transition metals but not gold and mercury

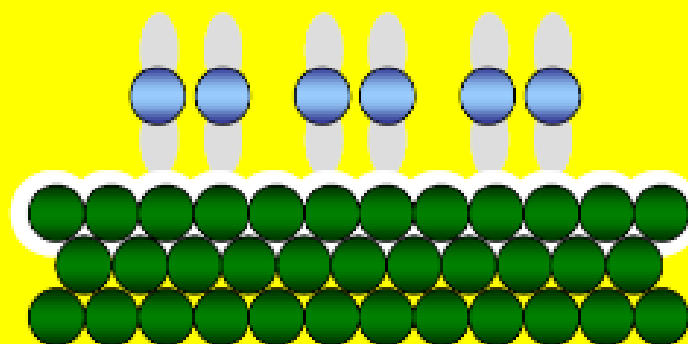
$$\Delta H_{\text{ads}} = 50\text{--}500 \text{ kJ/mole}$$

Can be activated, in which case equilibrium can be slow and increasing temperature can favour adsorption

Surface reactions may take place: dissociation, reconstruction, catalysis

MONOLAYER ADSORPTION

Langmuir isotherm used to model equilibrium



Factors Affecting Adsorption

- **Temperature:**

- Adsorption is an **exothermic** process.
- Therefore, in accordance with Le Chatelier's principle, the magnitude of adsorption should increase with decrease in temperature. It is in the case of physical adsorption.
- Chemical adsorption first increases with rise in the temperature and then starts decreasing.

Factors Affecting Adsorption

- **Pressure:**

- With increase of pressure, adsorption increases up to certain extent till saturation level is achieved - no more adsorption takes place no matter how high the pressure applied.

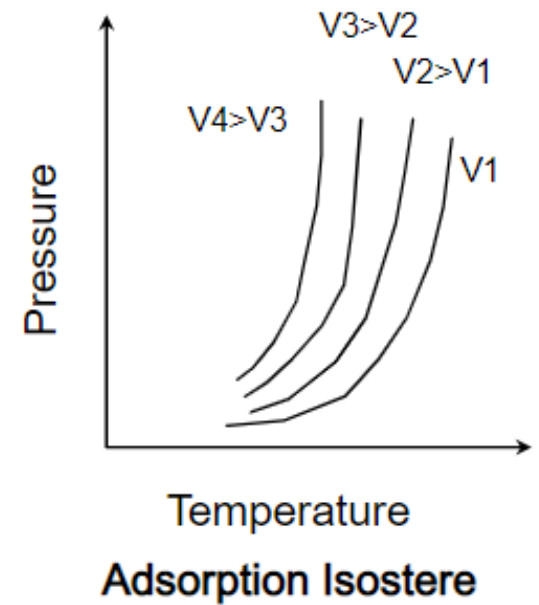
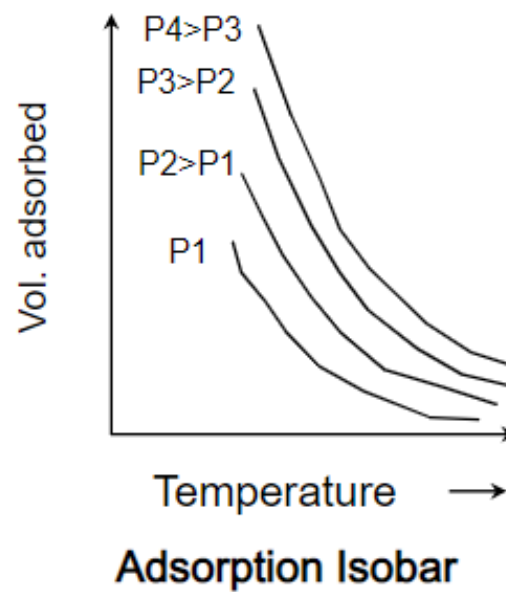
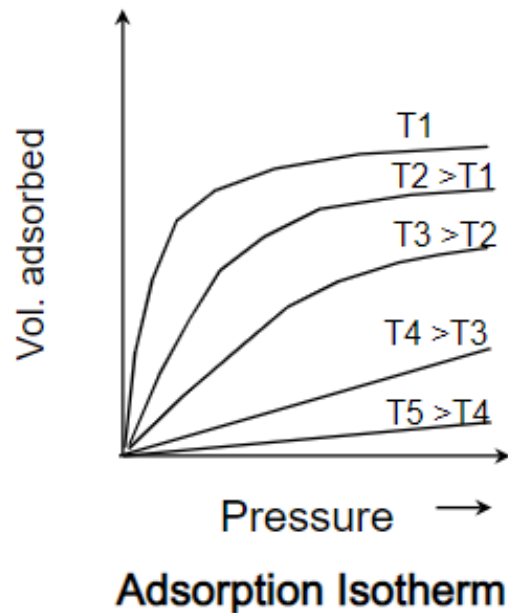
- **Surface area:**

- It's a surface phenomenon therefore adsorption capacity of adsorbent increases with increase in its surface area.
- Smaller the size of particles of solid adsorbents more is the extent of adsorption at its surface interface.

Characterisation of Adsorption

- ❑ **Adsorption isotherm:** It is a graph that expresses amount of adsorption as a function of pressure, at constant temperature.
- ❑ **Adsorption isobar:** It is a graph that expresses amount of adsorption as a function of temperature, at constant pressure.
- ❑ **Adsorption isostere:** It is a graph that expresses amount of adsorption as a function of temperature, at constant volume.

Characterisation of Adsorption



Adsorption isotherm

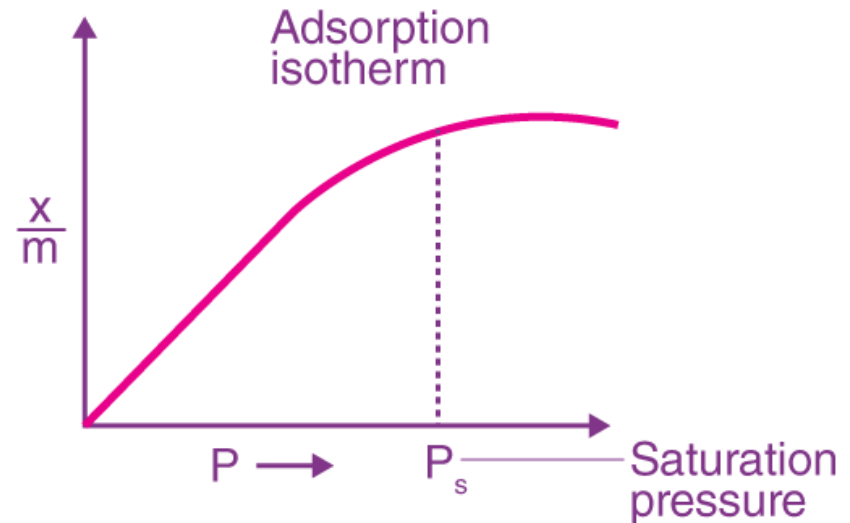
- An adsorption isotherm is a graph that represents the variation in the amount of adsorbate(x) adsorbed on the surface of the adsorbent with the change in pressure at a constant temperature.
- Different adsorption isotherms:

1. *Freundlich isotherm*

2. *Langmuir isotherm*

3. *Temkin Isotherm*

4. *BET Isotherm*



Adsorption Isotherms

- Plot of the amount of adsorbate on the adsorbent as a function of its pressure (if gas) or concentration (if liquid) at constant temperature.
- **Langmuir isotherm** (adsorbed layer one molecule thick)
- **Freundlich isotherm** (Heterogeneous adsorbent surface with different adsorption sites)
- **Brunauer, Emmett and Teller (BET) isotherm** (molecules can be adsorbed more than one layer thick)

Freundlich Adsorption Isotherm

- In 1909, German scientist Freundlich provided an empirical relationship between the amount of gas adsorbed by a unit mass of solid adsorbent and pressure at a particular temperature.

$$x/m = k.P^{1/n} \quad (n > 1)$$

$$\log x/m = \log k + 1/n \log P$$

Where, **x** = mass of the gas adsorbed

m = mass of the adsorbent

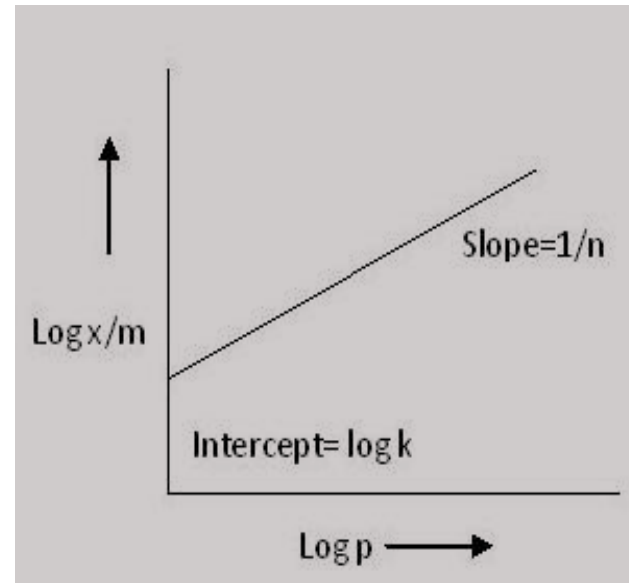
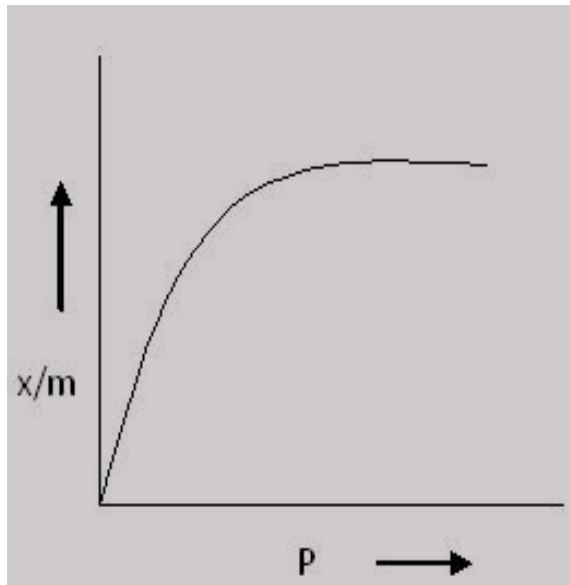
P = Pressure

‘**k**’ and ‘**n**’ (>1) are constants that depend on the nature of the adsorbent and the gas at a particular temperature.

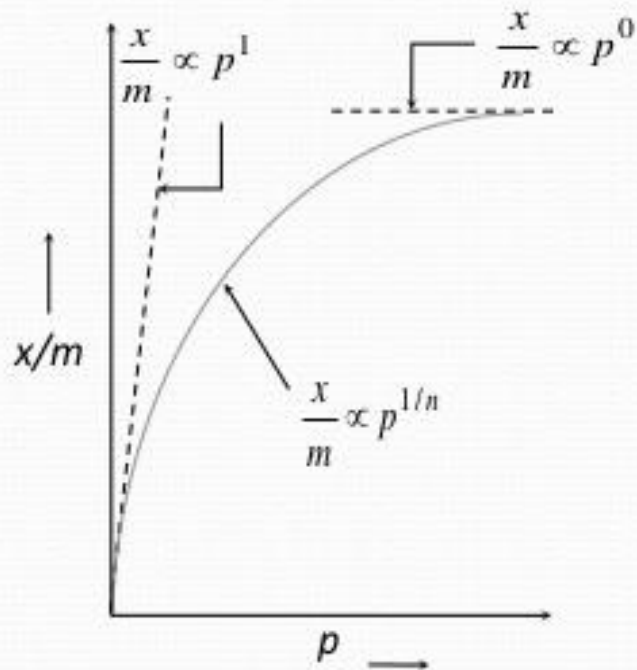
Freundlich Adsorption Isotherm

$$x/m = k.P^{1/n} \quad (n > 1)$$

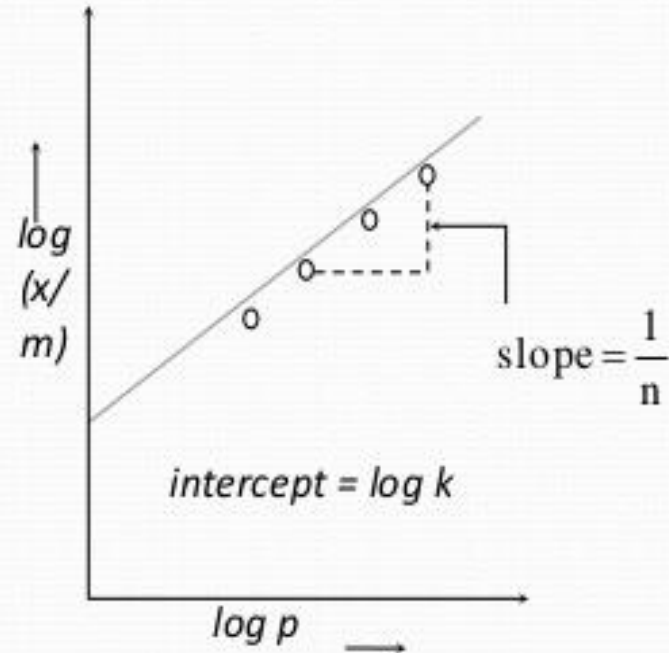
$$\log x/m = \log k + 1/n \log P$$



Freundlich Adsorption Isotherm



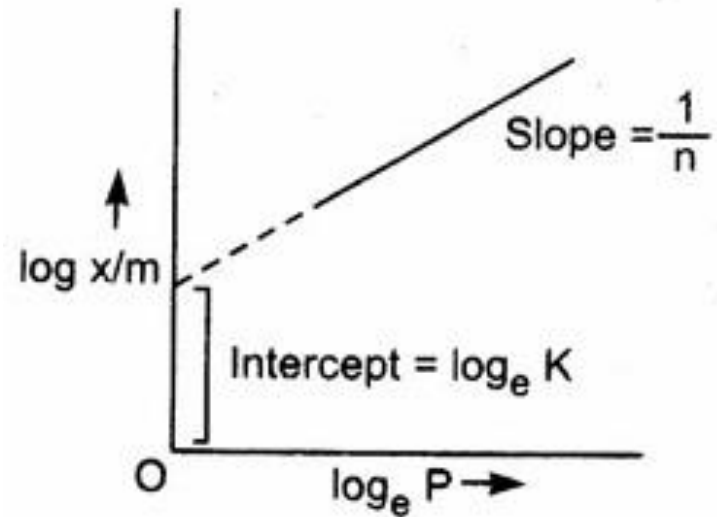
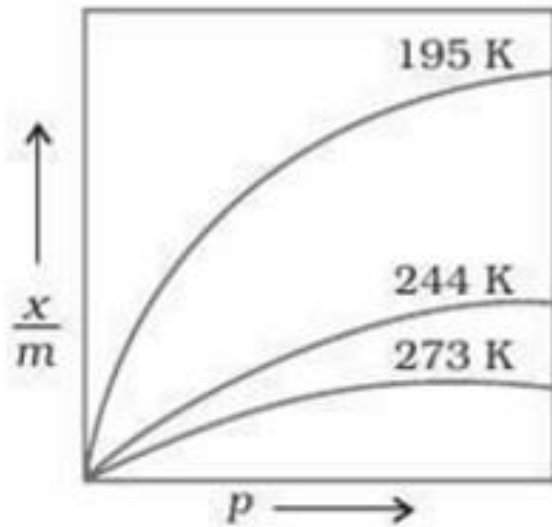
Freundlich adsorption isotherm: plot of x/m against p



Plot of $\log x/m$ against $\log p$ for the adsorption of a gas on a solid

This model is *inadequate to explain some cases of adsorption*.
In certain cases the adsorption continues even after the point of Freundlich saturation limit. (for example liquid condensation).

Freundlich Adsorption Isotherm



$$x/m = k.P^{1/n} \quad (n > 1)$$

$$\log x/m = \log k + 1/n \log P$$

Freundlich Adsorption Isotherm

$$x/m = k.P^{1/n} \quad (n > 1) \qquad \log x/m = \log k + 1/n \log P$$

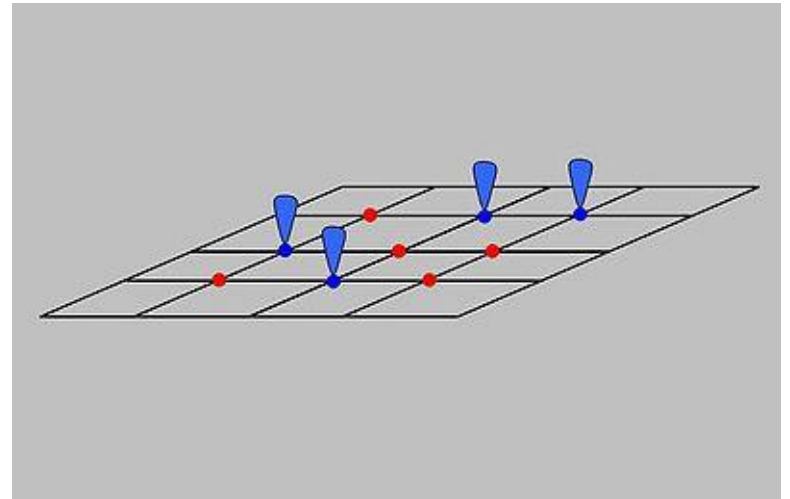
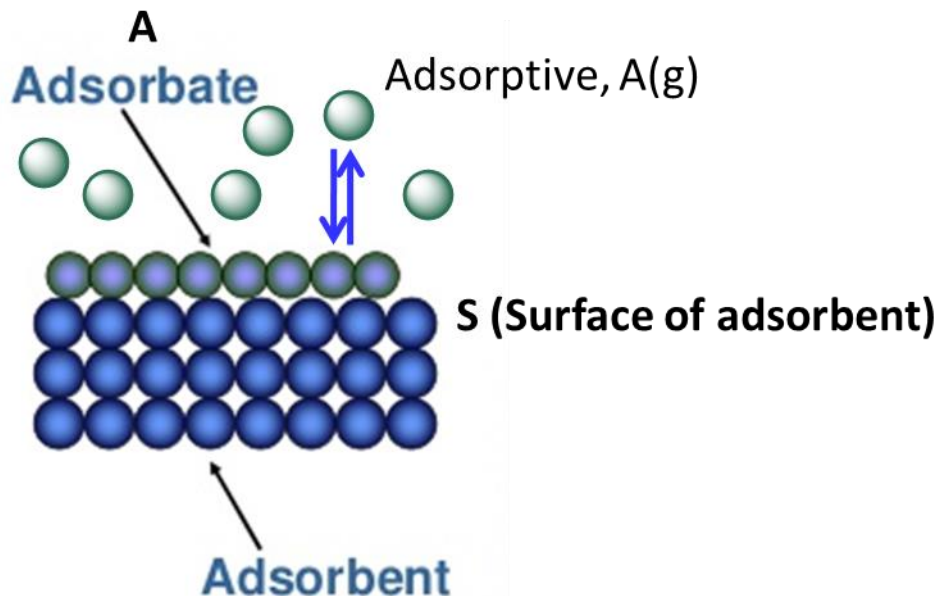
- ♦ When $1/n = 0$, $x/m = \text{constant}$, the adsorption is independent of pressure.
- ♦ When $1/n = 1$, $x/m = k p$, i.e. $x \propto p$, the adsorption varies directly with pressure.
- ♦ Both the conditions are supported by experimental results. **The experimental isotherms always seem to approach saturation at high pressure.**
- ♦ This cannot be explained by Freundlich isotherm. Thus, it fails at high pressure.

Freundlich Adsorption Isotherm

- ✓ It is based on empirical evidence.
- ✓ Multilayer adsorption
- ✓ Freundlich Isotherm correctly established the relationship of adsorption with pressure at lower values.
- **Limitation** : Failed to predict adsorption capacity at higher pressure

Langmuir Adsorption Isotherm

- In 1916, American scientist **Irving Langmuir** proposed another adsorption isotherm called Langmuir adsorption isotherm.



Langmuir Adsorption Isotherm

Assumptions of Langmuir Isotherm

1. Fixed number of **vacant** or **adsorption sites** are available on the surface of solid.
2. All the vacant sites are of **equal size and shape** on the surface of adsorbent.
3. Each site can hold maximum of **one gaseous** molecule and a constant amount of heat energy is released during this process.
4. Dynamic **equilibrium** exists between adsorbed gaseous molecules and the free gaseous molecules.



5. Adsorption is **monolayer or uni-layer**.

Langmuir Adsorption Isotherm

- Langmuir Equation depicts the relationship between the **number of active sites** of the surface undergoing adsorption (i.e. extent of adsorption) and pressure.

θ : fraction of sites which are covered with gaseous molecules.

$(1 - \theta)$: fraction of surface which are unoccupied by gaseous molecules.

$$\text{fraction of sites adsorbed} = \frac{\text{number of adsorption sites occupied}}{\text{number of adsorption sites available}}$$

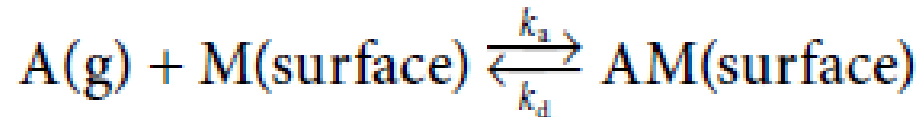
$$\text{Rate of adsorption} \propto P N (1 - \theta)$$

$$\text{Rate of desorption} \propto \theta N$$

$$\text{Rate of adsorption} = K_a P N (1 - \theta)$$

$$\text{Rate of desorption} = K_d \theta N$$

Langmuir Adsorption Isotherm



rate of adsorption $\frac{d\theta}{dt} = k_a p N (1 - \theta)$

rate of desorption $\frac{d\theta}{dt} = -k_d N \theta$

at equilibrium At equilibrium there is no net change in θ , implying that the sum of these two rates must be zero:

$$k_a p N (1 - \theta) - k_d N \theta = 0.$$

Langmuir Equation

Langmuir Adsorption Isotherm

$$k_a p N (1 - \theta) - k_d N \theta = 0.$$

solving for θ gives the **Langmuir isotherm**

$$\theta = \frac{Kp}{1 + Kp} \quad K = \frac{k_a}{k_d}$$

Langmuir Equation

Langmuir Adsorption Isotherm

$$\theta = \frac{KP}{(1 + KP)}$$

- **Case-I:**

At **lower pressure**, KP is so small that factor $(1 + KP)$ in denominator can almost be ignored

Langmuir equation reduces to : **$\theta = KP$**

- **Case-II:**

At **higher pressure**, KP is so large that factor $(1 + KP)$ in denominator is nearly equal to KP

Langmuir equation reduces to : **$\theta = Kp / Kp = 1$**

- **Limitation : Valid at low Pressures only**

Langmuir Adsorption Isotherm

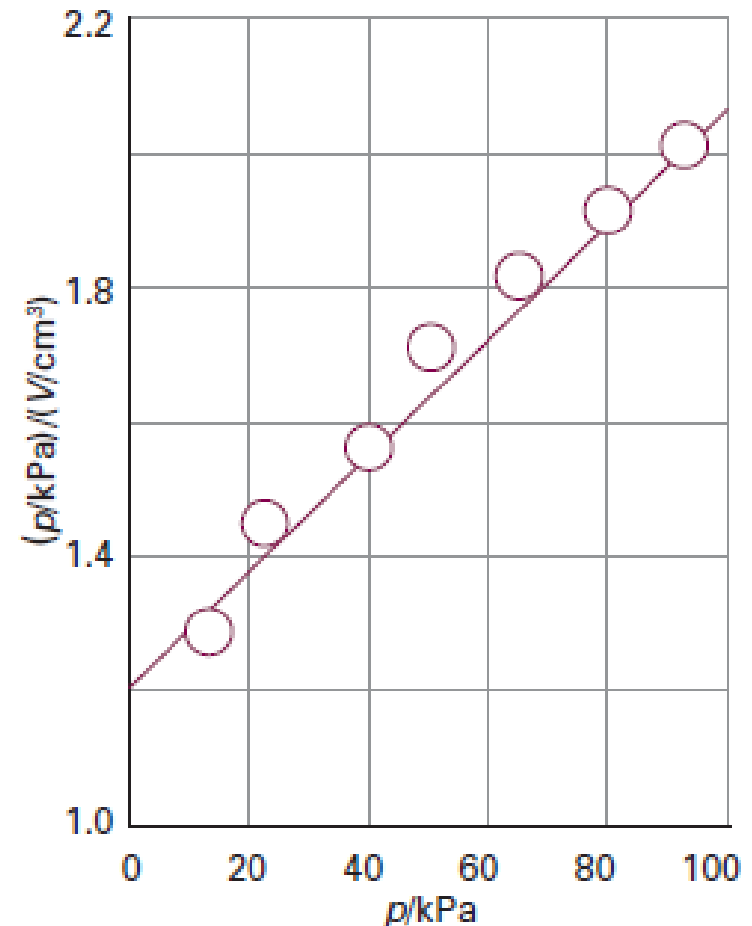
$$\theta = \frac{Kp}{1 + Kp}$$

$$\theta (1 + Kp) = Kp$$

$$\theta + \theta Kp = Kp$$

$$\theta = \frac{V}{V_{\infty}} \quad V_{\infty} : \text{Volume of complete coverage}$$

$$\frac{p}{V} = \frac{p}{V_{\infty}} + \frac{1}{KV_{\infty}}$$



Hence, a plot of p/V against p should give a straight line of slope $1/V_{\infty}$ and intercept $1/KV_{\infty}$.

Differences between **Freundlich** and **Langmuir** adsorption isotherms



FREUNDLICH ADSORPTION ISOTHERM

- Tells about the quantity of gas adsorbed by unit mass of solid adsorbent with pressure.
- Represented by formula $x/m = KP^{1/n}$



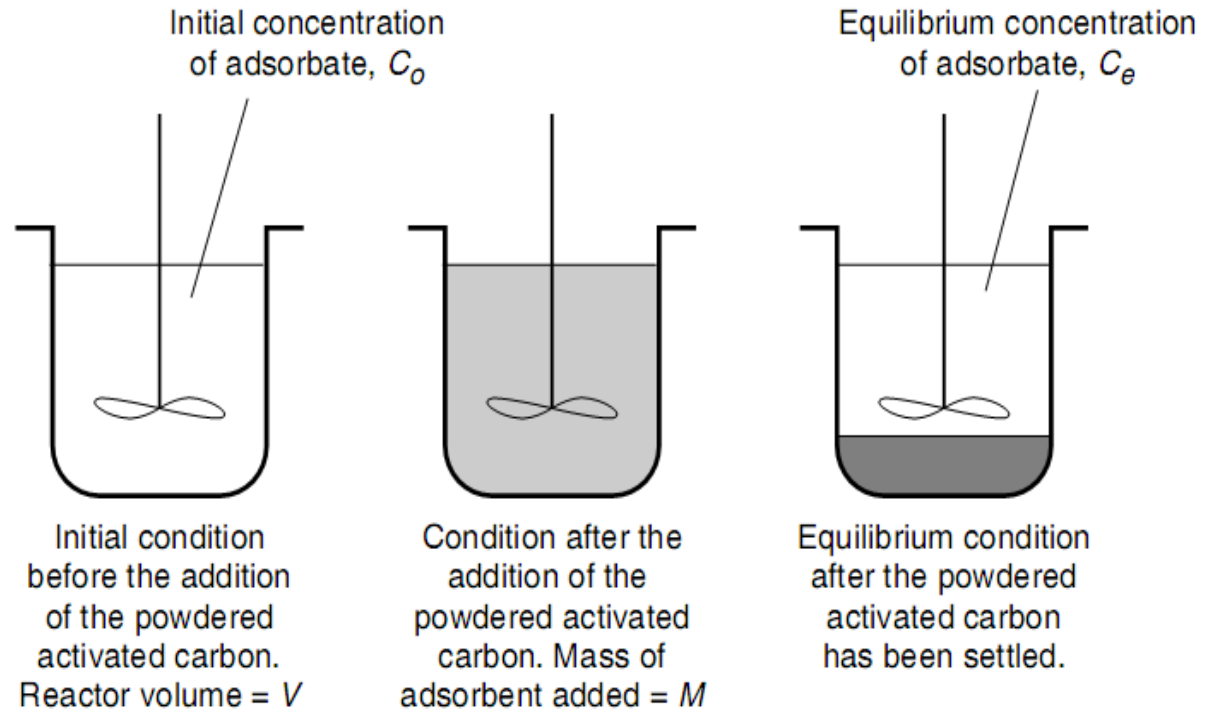
LANGMUIR ADSORPTION ISOTHERM

- Tells about the number of active site undergoing adsorption and pressure.
- Represented by formula $\theta = \frac{KP}{1 + KP}$

Adsorption isotherms: Solid - Liquid interaction

mass balance equation
for adsorption

$$q_e = \frac{(C_o - C_e)V}{m}$$



where

q_e = mass of material adsorbed (at equilibrium) per mass of adsorbent,
mg adsorbate/g adsorbent

C_o = initial concentration of adsorbate, **mg/L**

C_e = equilibrium concentration in solution (amount adsorbed = q_e , mg/L)

V = volume of liquid in the reactor, L

m = mass of adsorbent, g

Adsorption isotherms:

Solid - Liquid interaction

Adsorption isotherms: **Freundlich isotherm**

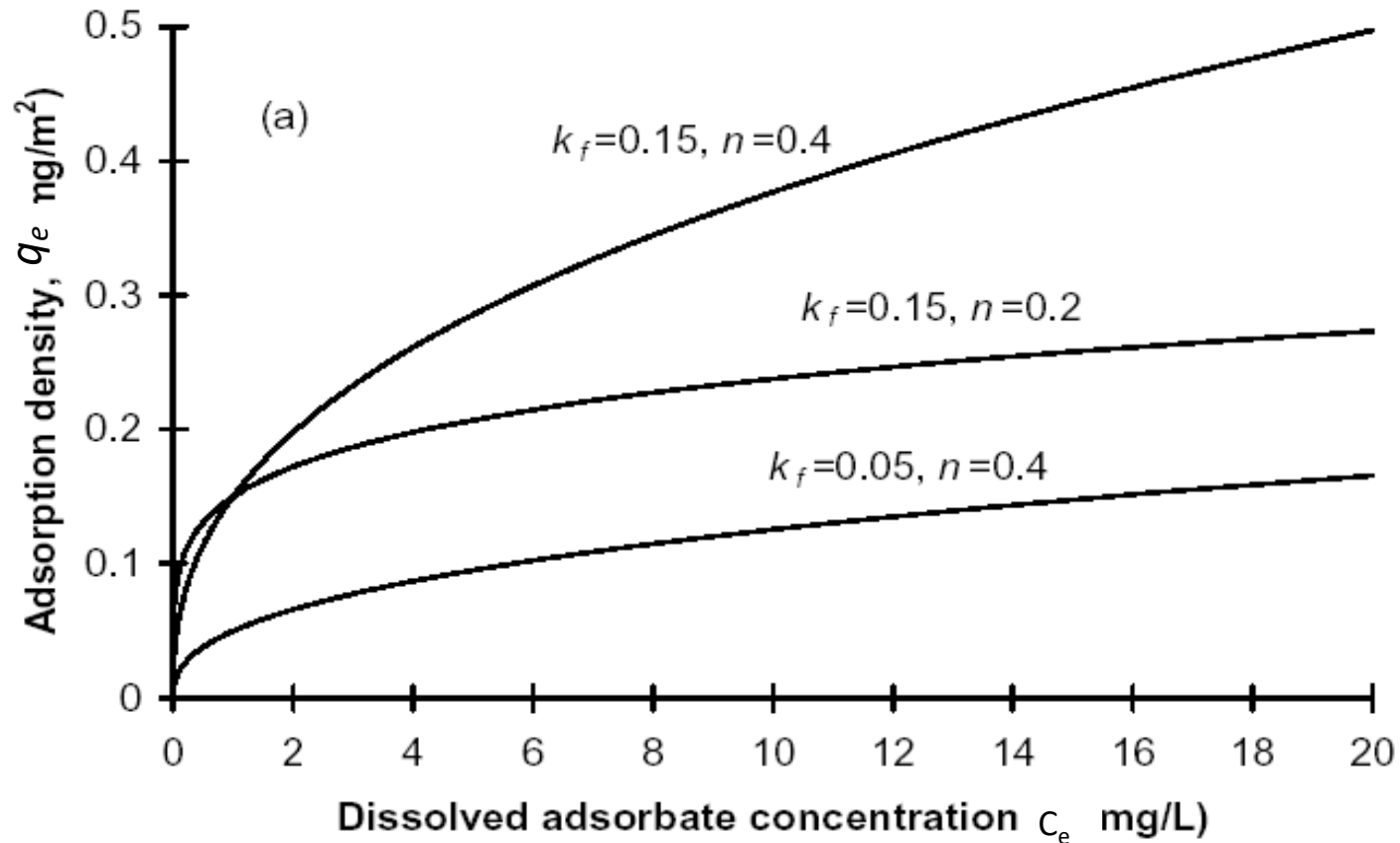
For the special case that the surface energies are heterogeneous (particularly good for mixed adsorbates) in which the energy term, " **k_f** ", varies as a function of surface coverage, the Freundlich model is appropriate.

The model equation is: $q_e = k_f C_e^{\frac{1}{n}}$

k_f & n are system specific constants

Shape of the **Freundlich isotherm changes** upon changing the values of **n** and **k_f**

$$q_e = k_f C_e^{\frac{1}{n}}$$

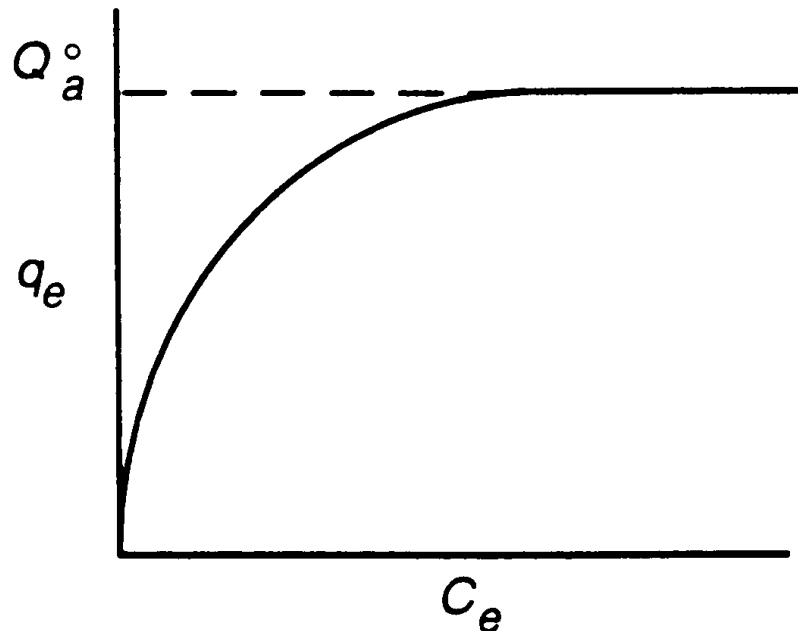


Adsorption isotherms:

Solid - Liquid interaction

Adsorption isotherms: **Langmuir model**

This model assumes **monolayer** coverage and constant binding energy between surface and adsorbate.



$$q_e = \frac{K \cdot Q_a^0 \cdot C_e}{1 + K \cdot C_e}$$

$$\frac{C_e}{q_e} = \frac{1}{K Q_a^0} + \frac{C_e}{Q_a^0}$$

Q_a^0 : the maximum adsorption capacity
(monolayer coverage) (g solute/g adsorbent).

K Langmuir equilibrium constant (L/mg)

Adsorption isotherms: Solid-liquid interaction

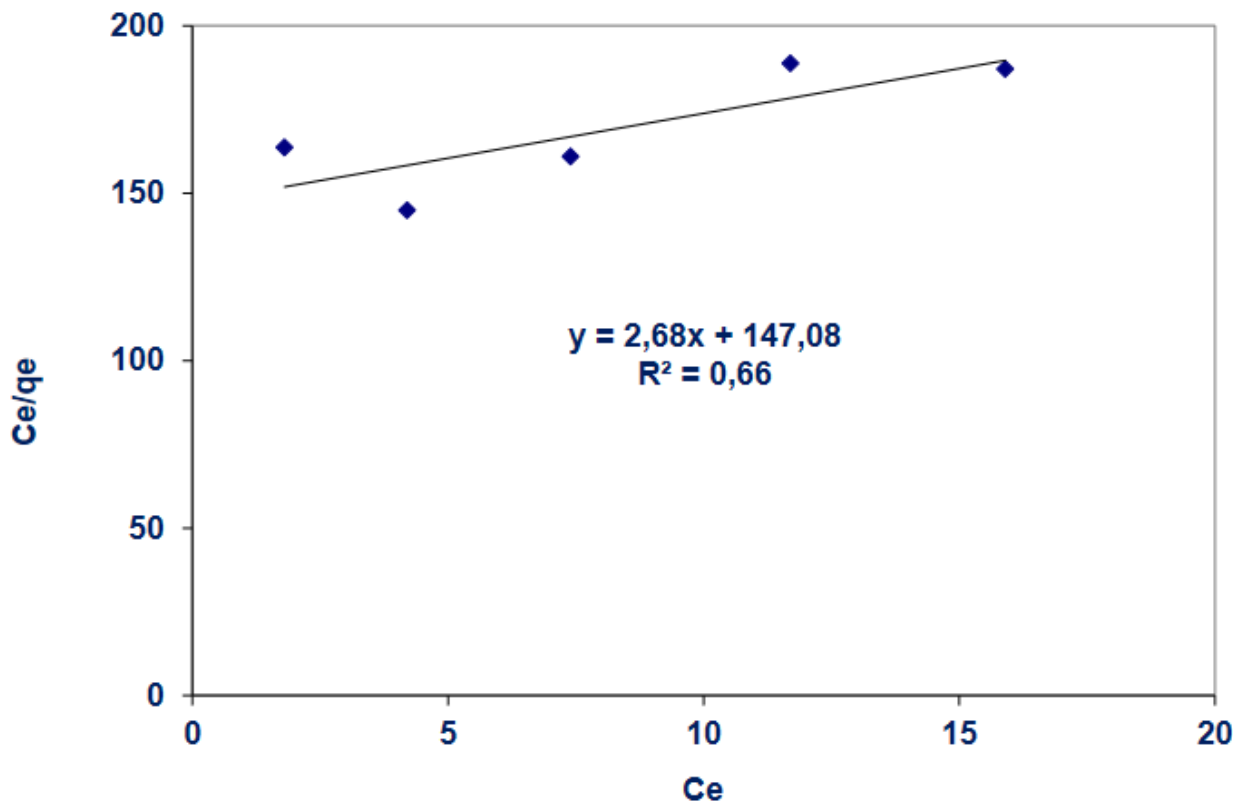
Adsorption isotherms: Langmuir model

$$q_e = \frac{K \cdot Q_a^0 \cdot C_e}{1 + K \cdot C_e}$$

$$\frac{C_e}{q_e} = \frac{1}{K Q_a^0} + \frac{C_e}{Q_a^0}$$

$$\text{Slope} = \frac{1}{Q_a^0}$$

$$\text{Intercept} = \frac{1}{K Q_a^0}$$



Application of Adsorbents

- Organic contaminants such as *pharmaceuticals, cosmetics, pesticides* can be removed from water using the method of **adsorption**.
- To speed up chemical reactions (catalysis)
- To remove bad odor and color in water
- Used as a solid support purify proteins and chemical compounds
- Used to remove unwanted impurities from gas

Uses of activated carbon as a adsorbent

- Uses

- Remove man-made organic chemicals

- Remove miscellaneous tastes and odor from water **assuming no bacterial problems**

- Remove radon gas from water

- Maintenance

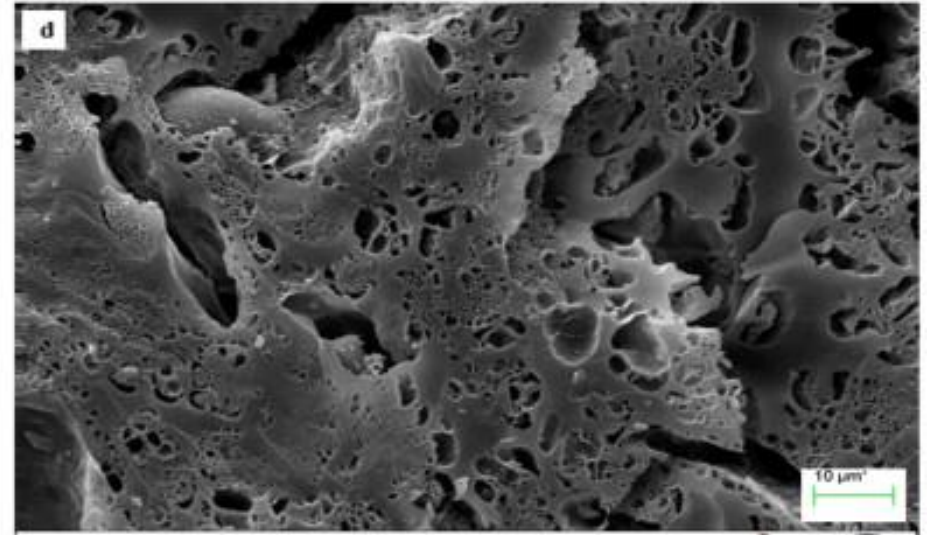
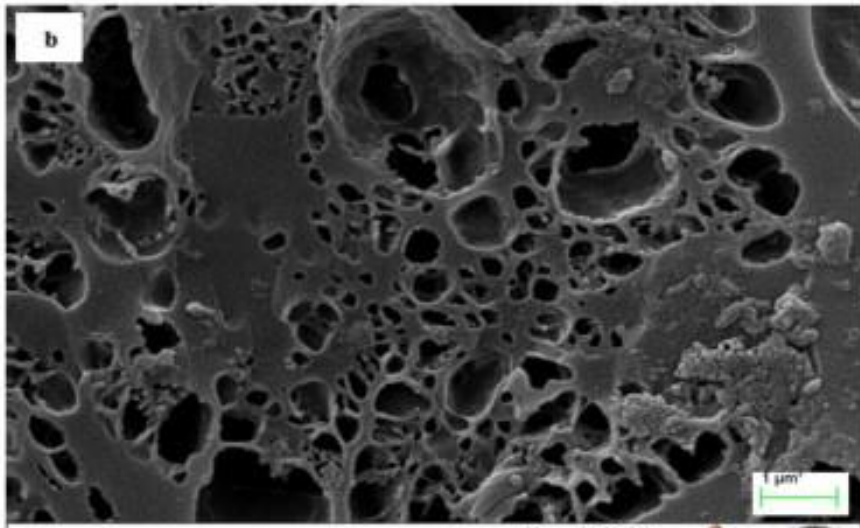
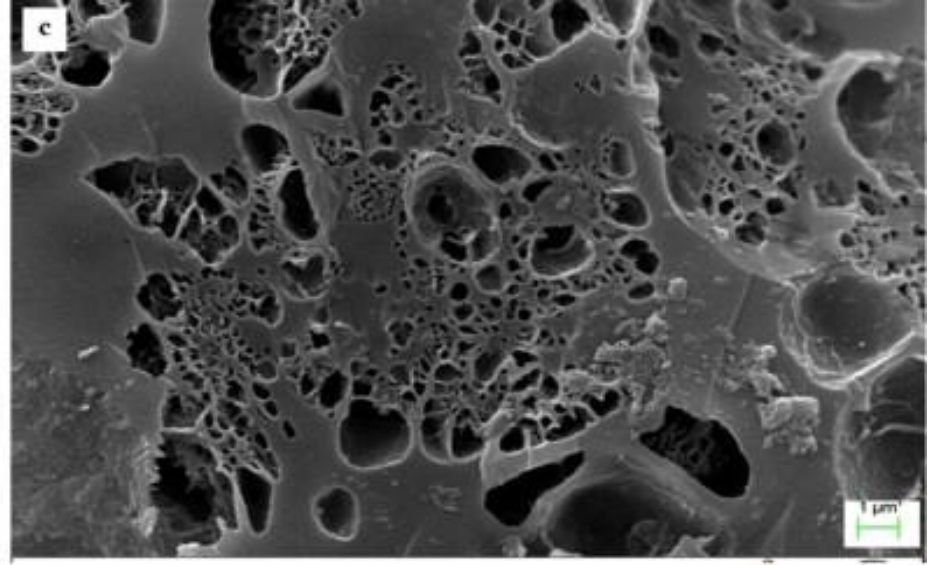
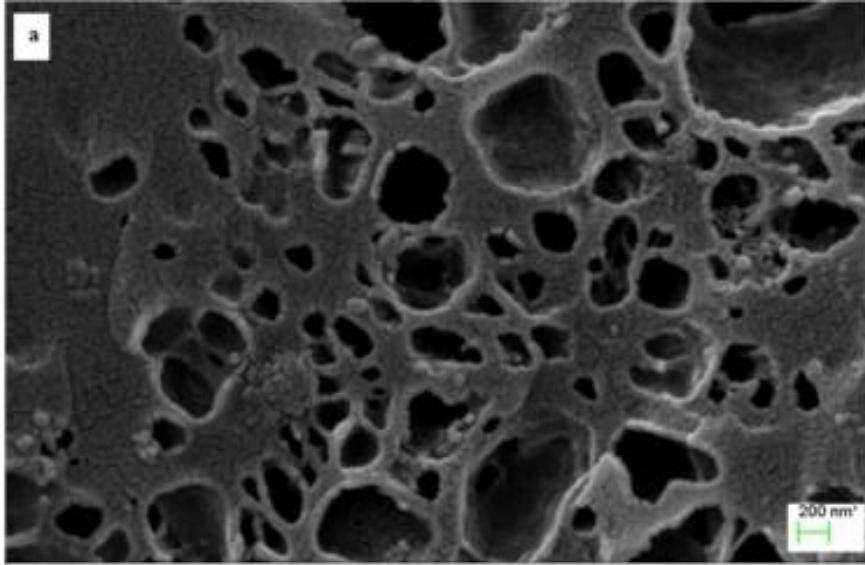
- Carbon must be replaced routinely





Porous Nature of Activated Carbon

Generally activated carbon is porous. They provide large surface area (m^2/g). Pore size depends on the carbon source.



Problems

Example 1

Adsorption of benzene onto activated carbon has been reported to obey the following Freundlich isotherm equation, where c is in mg/L and q is in mg/g:

$$q_{\text{benz}} = 50.1 c_{\text{benz}}^{0.533}$$

A solution at 25°C containing 0.50 mg/L benzene is to be treated in a batch process to reduce the concentration to less than 0.01 mg/L. The adsorbent is activated carbon with a specific surface area of 650 m²/g.

Compute the required activated carbon dose.

Example 1

Solution:

$$q_e = \frac{(C_o - C_e)V}{m}$$

$$C_o = 0.5 \text{ mg/L}$$

$$C_e = 0.01 \text{ mg/L}$$

$$q_e = ?$$

$$\frac{m}{V} = ?$$

$$q_e = k C_e^{1/n}$$

$$q_{\text{benz}} = 50.1 C_{\text{benz}}^{0.533}$$

$$q_e = 50.1 (0.01)^{0.533}$$

$$q_e = 4.30 \text{ mg/g}$$

Example 1

Solution:

$$q_e = \frac{(C_o - C_e)V}{m}$$

$$C_o = 0.5 \text{ mg/L}$$

$$C_e = 0.01 \text{ mg/L}$$

$$q_e = 4.30 \text{ mg/g}$$

$$\frac{m}{V} = ?$$

$$\frac{m}{v} = \frac{C_o - C_e}{q_e}$$

$$\frac{m}{v} = \frac{0.5 - 0.01}{4.3} \text{ mg/L}$$

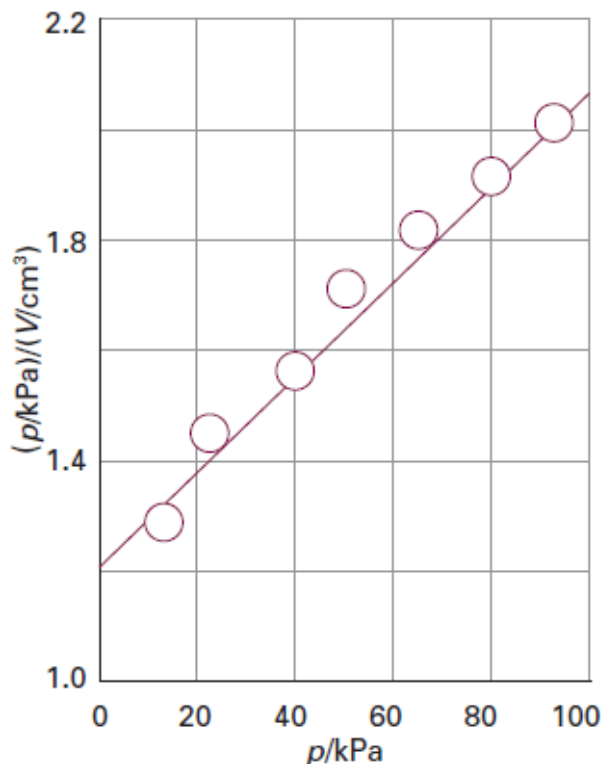
$$\frac{m}{v} = 0.114 \text{ g/L}$$

Example 2

The data given below are for the adsorption of CO on charcoal at 273 K. Confirm that they fit the Langmuir isotherm, and find the constant K and the volume corresponding to complete coverage.

p/kPa	13.3	26.7	40.0	53.3	66.7	80.0	93.3
V/cm^3	10.2	18.6	25.5	31.5	36.9	41.6	46.1

Solution:



a plot of p/V against p should give a straight line of slope $1/V_\infty$ and intercept $1/KV_\infty$.

$$\frac{P}{V_a} = \frac{1}{V_m K} + \frac{P}{V_m}$$

$$1/V_m = 1/V_\infty$$

where

V_a = volume of gas adsorbed at pressure P ;

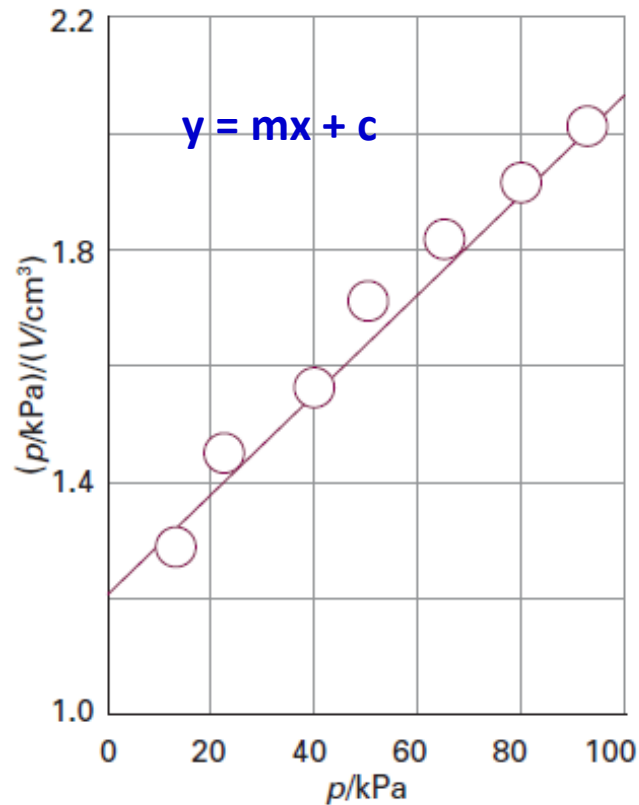
V_m = volume of gas required to form monolayer;

K = equilibrium constant and

P = pressure of adsorbate.

Example 2

Solution:



$$\text{Slope} = \frac{1}{V_{\infty}} = 0.00900 \text{ (from graph)}$$

$$V_{\infty} = 111 \text{ cm}^3$$

$$\text{Intercept} = \frac{1}{KV_{\infty}} = 1.20 \text{ (from graph)}$$

$$K = \frac{1}{(111 \text{ cm}^3) \times (1.20 \text{ kPa cm}^{-3})} = 7.51 \times 10^{-3} \text{ kPa}^{-1}$$

Example 3

Iron – coated clays (“red clays”) can adsorb phosphorus. A municipality is considering the use of these materials for the removal of P from the town’s wastewater prior to discharge into a lake. The following adsorption data are obtained in the laboratory using the clay and secondary effluent from the town’s treatment facility.

q_e P adsorbed, mg P / g clay	C_e Equilibrium P in solution, mg P / L
0.2	0.055
0.3	0.088
0.5	0.182
1.0	0.50
1.7	2.78

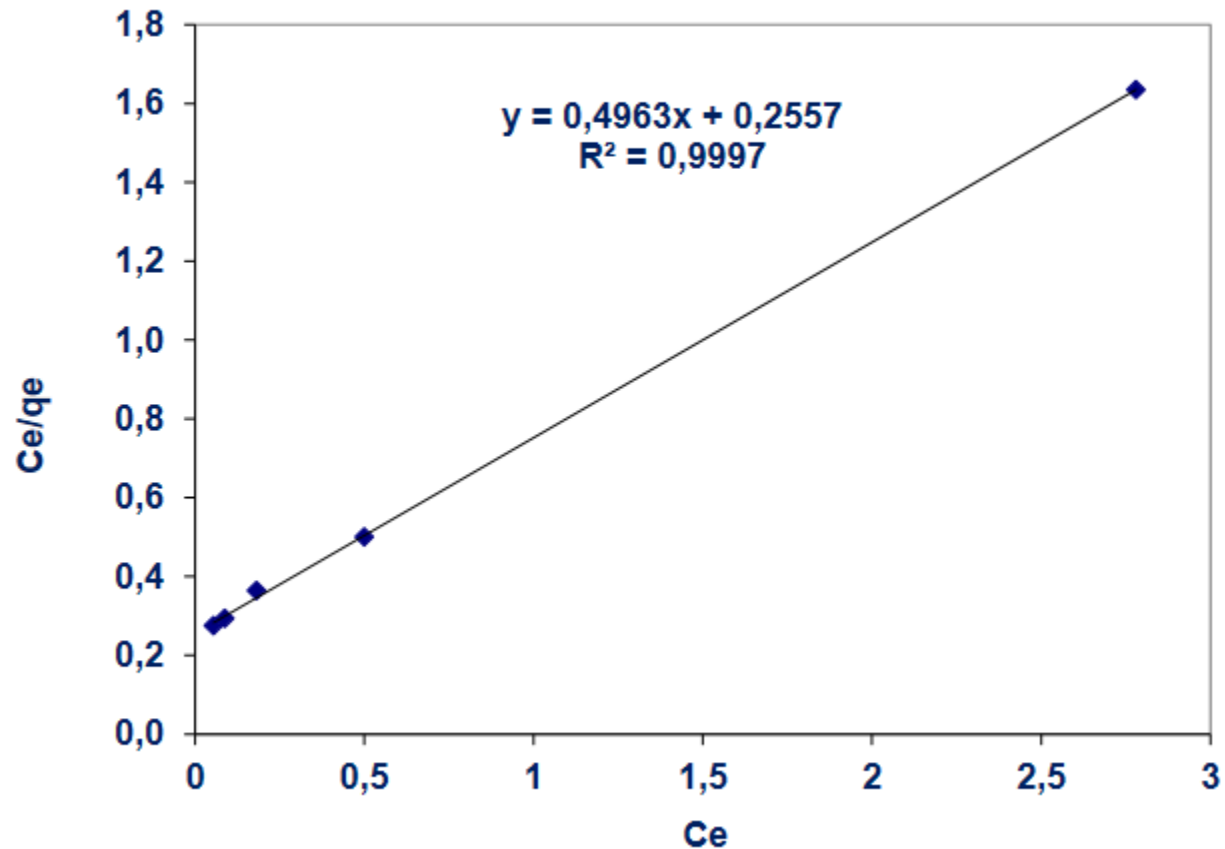
Determine the maximum P adsorption capacity of this clay (mg P / g clay).

Example 3

C_e/q_e
0.275
0.293
0.364
0.5
1.635

Solution:

$$\frac{C_e}{q_e} = \frac{1}{K Q_a^0} + \frac{C_e}{Q_a^0}$$



Slope of (C_e/q_e) versus C_e curve = $\frac{1}{Q^0} = 0.496$

$$Q^0 = 2.02 \text{ mg P / g clay}$$

